

Coincidences and Growth of Boxed Mesh Patterns

Sergey Kitaev¹, Dun Qiu², and Chao Xu²

¹Department of Mathematics and Statistics, University of Strathclyde,
Glasgow G1 1XH, United Kingdom
`sergey.kitaev@strath.ac.uk`

²Center for Combinatorics, LPMC, Nankai University,
Tianjin 300071, P. R. China
`qiudun@nankai.edu.cn, xclyon1@gmail.com`

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Abstract

A boxed mesh pattern is a mesh pattern whose selected entries lie in an empty axis-parallel rectangle. We classify coincidences of boxed patterns with classical and vincular patterns, exhibit a genuinely bivincular coincidence, and prove that no boxed-bivincular coincidence occurs for patterns of length at least five. Together with known results, this shows that every boxed pattern of length at least five has factorial growth and hence fails the Stanley–Wilf property. At length four, one exceptional orbit is enumerated by the semi-Baxter numbers, while the remaining exceptional orbit, $\{2143, 3412\}$, is unresolved; we conjecture that it has factorial growth.

For $\text{Box}(123)$, we derive an exact maximum-insertion identity and prove the subfactorial upper bound $2^{5n}n^{\beta n}$, where $\beta = \log_2(2\cos(\pi/7)) < 0.85$. A closed enumeration remains open. We also prove a general first-moment formula for boxed mesh patterns that depends only on the length of the underlying pattern; in particular, the expected number of $\text{Box}(123)$ occurrences in a uniformly random permutation of length n is asymptotic to $n \log n$.

For $\text{Box}(12)$, we identify the occurrence statistic with the up-degree in the strong Bruhat order, obtaining its maximum, its mean, and an exact insertion identity for the distribution polynomials. We conjecture that the coefficients of these polynomials are unimodal.

Keywords. mesh pattern; boxed pattern; Stanley–Wilf property; vincular pattern; bivincular pattern; semi-Baxter permutation; strong Bruhat order.

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1 Introduction

Let $\mathfrak{S}_n$ denote the set of permutations of $[n] = \{1, 2, \dots, n\}$, with $\mathfrak{S}_0 = \{\varepsilon\}$. A *classical pattern* of length k is a permutation $p = p_1p_2 \cdots p_k \in \mathfrak{S}_k$. An occurrence of p in a permutation $\pi = \pi_1\pi_2 \cdots \pi_n \in \mathfrak{S}_n$ is a subsequence $\pi_{i_1}\pi_{i_2} \cdots \pi_{i_k}$, where $i_1 < i_2 < \cdots < i_k$, that is order-isomorphic to p ; that is, $\pi_{i_r} < \pi_{i_s}$ if and only if $p_r < p_s$ for $1 \leq r, s \leq k$. For example, the pattern 132 occurs twice in the permutation 24513, namely as the subsequences 243 and 253.

For a pattern p , defined in any specified sense, let $\text{Av}_n(p) \subseteq \mathfrak{S}_n$ denote the set of permutations that avoid p , that is, contain no occurrence of p . The *Stanley–Wilf limit* of a pattern p , when it exists, is

$$L(p) = \lim_{n \rightarrow \infty} |\text{Av}_n(p)|^{1/n}.$$

Arratia [2] proved that an exponential upper bound for $|\text{Av}_n(p)|$ for a classical pattern p implies the existence of this limit, and Marcus and Tardos [25] proved the required upper bound. Thus the Stanley–Wilf theorem says that $L(p)$ exists and is finite for every fixed classical pattern p . For any pattern p , not necessarily classical, we shall say that the *Stanley–Wilf property* holds if

$$\limsup_{n \rightarrow \infty} |\text{Av}_n(p)|^{1/n} < \infty.$$

We say that a positive sequence (a_n) has *factorial growth* if $\log a_n = \Theta(n \log n)$, or, equivalently, if $a_n = n^{\Theta(n)}$. If p is a consecutive pattern of length at least three, meaning that the entries in an occurrence must occupy consecutive positions, Elizalde [14, Theorem 4.1] proved that there exist constants $0 < c_p < d_p < 1$ such that, for all sufficiently large n ,

$$c_p^n n! < |\text{Av}_n(p)| < d_p^n n!.$$

Consequently its n th root tends to infinity. (Consecutive patterns of length two are the exceptional trivial cases, with one avoider.) Vincular patterns, defined in Section 2, already exhibit both types of behaviour in length three: for example, $\underline{123}$, also written 1-23, has the Bell numbers as its avoidance sequence, whereas some other one-bond patterns have Catalan avoidance sequences [11]. Since $B_n^{1/n} \sim n/(e \log n)$, Bell growth also violates the Stanley–Wilf property.

Mesh patterns, defined in Section 2, were introduced by Brändén and Claesson [10] as a flexible language for imposing geometric conditions on occurrences of classical permutation patterns. They have since developed in several complementary directions: shading and coincidence criteria [29, 13], systematic and algorithmic classification [18, 5], exhaustive generation [17], asymptotic containment [16], and the study of distributions [22, 23, 24].

Two patterns are called *coincident* when they impose exactly the same avoidance condition: every permutation avoids one if and only if it avoids the other. Thus coincidence identifies the actual avoidance classes, not merely their counting sequences; a formal definition is given in Section 2.

Avgrustinovich, Kitaev and Valyuzhenich [3] introduced *boxed patterns*: all $(k-1)^2$ internal cells of a mesh pattern of length k are shaded. They showed, among other things, that the permutations avoiding boxed 132 are exactly those avoiding classical 132 and are therefore counted by the Catalan numbers. They also showed that an analogue of the Erdős–Szekeres theorem fails for boxed patterns of length greater than two, and they enumerated permutations avoiding two or more boxed patterns of length three, where generalized Catalan numbers appear. More strikingly, they proved factorial lower bounds for many boxed patterns, showing that the Stanley–Wilf conclusion for classical patterns does not extend to arbitrary mesh patterns. The simplest of their constructions doubles each letter:

$$\delta(\pi_1 \cdots \pi_m) = (2\pi_1)(2\pi_1 - 1) \cdots (2\pi_m)(2\pi_m - 1).$$

Every pair is a descending pair of consecutive values, and the empty-rectangle condition shows that $\delta(\pi)$ avoids $\text{Box}(123)$: the unselected sibling of the middle member of any increasing triple lies inside its rectangle. This injects $\mathfrak{S}_m$ into $\text{Av}_{2m}(\text{Box}(123))$. The same observation and its complement handle any boxed pattern having a monotone interval

factor of length three. More generally, factorial growth is inherited from any boxed interval factor; see Proposition 3.4.

Their general proof calls $\tau = \tau_1 \cdots \tau_k$ *good* when some $1 < i < k$ has $1 < \tau_i < k$ and the factor $\tau_{i-1}\tau_i\tau_{i+1}$ is nonmonotone. In the representative 213 case, every letter of an arbitrary base permutation is replaced by an increasing pair of consecutive values. Whichever sibling plays τ_i , its mate either lies in the forbidden rectangle or forces the adjacent member of the pattern to be the mate, contradicting the required order. Reverse and complement deal with the other three nonmonotone triples. If τ is not good, its entries split into monotone blocks around its minimum and maximum. Either it has a monotone factor of length three, in which case the Box(123) or Box(321) construction applies, or it is one of the four exceptional permutations below. Together these ideas give factorial lower bounds for twenty of the twenty-four underlying permutations of length four. Their argument left

$$2143, \quad 3412, \quad 2413, \quad 3142, \quad (1)$$

and treated them as a single symmetry class.

It is important that the same theorem already settles every larger length. More precisely, [3, Theorem 3] proves

$$|\text{Av}_n(\text{Box}(\tau))| \geq \left\lfloor \frac{n}{2} \right\rfloor !$$

for every $\tau \in \mathfrak{S}_k$, $k \geq 4$, other than the four permutations in (1). Consequently every boxed pattern of length at least five fails the Stanley–Wilf property. Thus the obstruction to coincidence with a classical, vincular or bivincular pattern in lengths at least five is accompanied by a much stronger growth statement, but the two facts have logically independent proofs: noncoincidence alone would not imply factorial growth.

Three questions from the final section of [3] are directly relevant here: determine the Stanley–Wilf behaviour of the four patterns in (1); enumerate Box(123); and decide when a boxed pattern can be coincident with a classical pattern. The last question has since been settled: coincident mesh patterns have the same underlying permutation [13, Lemma 4.1], and the enclosed-diagonal criterion of Tenner [29] classifies when a mesh is superfluous. For boxed patterns the complete list is recalled in Proposition 5.1. The first two questions remain the main enumerative motivation for this paper.

Boxed patterns are also called *frame patterns*; under that name their occurrence statistics have been studied in the cycle structure of permutations by Jones, Kitaev and Remmel [19]. On the algorithmic side, deciding boxed containment has been studied by Bruner and Lackner [6], by Cho, Na and Sim [12] and by Amit, Bille, Hagge Cording, Li Gørtz and Wedel Vildhøj [4]; see Remark 3.3.

The four exceptional cases separate in two ways. First, the four patterns in (1) are treated together in [3, p. 48], but they do not form a single symmetry class: reverse, complement and inverse split (1) into

$$\{2143, 3412\} \quad \text{and} \quad \{2413, 3142\}. \quad (2)$$

The two avoidance sequences are different: at size five they have respectively 106 and 104 elements. Second, the latter orbit is not a counterexample to the Stanley–Wilf property. The key observation for this orbit, that we prove in Lemma 4.1, is the coincidence

$$\text{Box}(2413) \equiv_c 2\underline{4}13,$$

where the underline requires the entries playing 4 and 1 to be adjacent; the equivalent dash notation is 2-41-3. The right-hand class consists of the semi-Baxter permutations,

completely enumerated by Bouvel, Guerrini, Rechnitzer and Rinaldi [7]. This settles one of the two orbits in (2) and explains the value 104 at size five. We also classify all coincidences of a boxed pattern with a nonclassical vincular pattern: none exist in lengths five or more. Moreover, $\text{Box}(2413)$ is coincident with a genuinely bivincular pattern, and we prove that no boxed pattern of length at least five is coincident with any bivincular pattern.

For the unresolved orbit $\{2143, 3412\}$, we show that replacing every entry x by the increasing pair $(2x - 1)(2x)$ preserves the number of boxed 2143 occurrences. We also conjecture that the avoidance sequence associated with this orbit has factorial growth; see Conjectures 7.4 and 7.5 and the accompanying discussion in Section 7.

Our second focus is boxed 123. The sequence is recorded as OEIS A201168 [26], but no generating function or coefficient formula is given there or, to our knowledge, elsewhere in the literature. We give an exact maximum-insertion identity and show that its prefix condition is itself enumerated by $(t - 1)!$; this gives the five new terms $a_{13}, \dots, a_{17}$. A weighted halving argument also gives

$$|\text{Av}_n(\text{Box}(123))| \leq 2^{5n} n^{\beta n}, \quad \beta = \log_2 \left(2 \cos \frac{\pi}{7} \right) < 0.85,$$

so the avoidance class is subfactorial even though it does not have the Stanley–Wilf property. The identity does not close in the total numbers, and a closed enumeration remains open. On the distributional side, we prove that the expected number of occurrences of $\text{Box}(\tau)$ in a uniformly random permutation depends only on the length of τ , not on τ itself. For boxed 123 the exact mean is $(n + 2)H_n - 3n \sim n \log n$, so a random permutation contains on average on the order of $n \log n$ such occurrences.

A complementary distribution question concerns the boxed pattern 12. A recent sequence of papers has completed the classification of length-two mesh patterns by distribution-equivalence and Wilf-equivalence. Su, Kitaev and Zhang [28] obtained a near-classification; Fang, Fu, Kitaev, Li, Su and Sun [15] reduced both classifications to a single remaining case; and Zhang and Zhao [30] settled that case. The first of these papers records the initial distribution polynomials for boxed 12 but leaves a general formula open. We observe that the number of boxed 12 occurrences is exactly the up-degree of a permutation in the strong Bruhat order. The standard Bruhat cover criterion, also discussed in mesh-pattern language by Bouvel, Ferrari and Tenner [8], therefore brings the results of Adin and Roichman [1] into the problem. In particular, it gives the maximum number of occurrences, the number of maximizers, and the exact mean. We add a short direct proof of the mean and an insertion identity that isolates the catalytic prefix statistic needed for the full distribution.

The paper is organized as follows. In Section 2, we introduce the notation and recall the preliminary results used throughout the paper. In Section 3, we translate boxed containment into consecutive containment after restriction to an interval of values. In Section 4, we correct the length-four symmetry reduction, identify the semi-Baxter orbit, and record the resulting global growth classification. In Section 5, we classify classical and vincular coincidences, exhibit a genuinely bivincular coincidence, and prove that no boxed–bivincular coincidence occurs in length at least five. In Section 6, we give an exact identity, a subfactorial upper bound, and a general first-moment formula motivated by boxed 123. In Section 7, we discuss the remaining orbit and state two conjectures. Finally, in Section 8, we connect boxed 12 with covers in the strong Bruhat order and record the resulting distributional information.

2 Preliminaries

If $w = w_1 \cdots w_k$ is a word with distinct letters, its *standardization*, also called its *reduced form*, is the unique permutation $\text{st}(w) \in \mathfrak{S}_k$ whose entries have the same relative order as those of w . Recall that a permutation π *contains* the classical pattern $\tau \in \mathfrak{S}_k$ if $\text{st}(\pi_{i_1} \cdots \pi_{i_k}) = \tau$ for some $i_1 < \cdots < i_k$; otherwise, π avoids τ .

A *mesh pattern* of length k is a pair $P = (\tau, R)$, where $\tau \in \mathfrak{S}_k$ and $R \subseteq \{0, 1, \dots, k\}^2$. Its diagram consists of the plot of τ , with points (r, τ_r) for $1 \leq r \leq k$, together with a shaded unit square $[a, a+1] \times [b, b+1]$ for each $(a, b) \in R$. For example,

$$P = (2413, \{(0, 1), (2, 2), (3, 0)\}) = \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline \end{array}$$

Let $\pi = \pi_1 \cdots \pi_n \in \mathfrak{S}_n$, and suppose that $i_1 < \cdots < i_k$ determine a classical occurrence of τ in π . Let $v_1 < \cdots < v_k$ be the selected values $\pi_{i_1}, \dots, \pi_{i_k}$ written in increasing order, and set $i_0 = v_0 = 0$ and $i_{k+1} = v_{k+1} = n+1$. The selected entries form an *occurrence* of the mesh pattern $P = (\tau, R)$ if, for every $(a, b) \in R$, there is no plot point (j, π_j) satisfying $i_a < j < i_{a+1}$ and $v_b < \pi_j < v_{b+1}$. Thus each shaded square prescribes a rectangular region that must contain no other point of the plot of π . When $R = \emptyset$, this definition reduces to classical pattern containment.

The *boxed pattern* with underlying permutation $\tau \in \mathfrak{S}_k$ is $\text{Box}(\tau) = (\tau, \{1, \dots, k-1\}^2)$. It is also denoted by a box drawn around the permutation, such as $\boxed{123}$, or by displaying its mesh:

$$\text{Box}(123) = \boxed{123} = \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline & & \\ \hline \end{array}$$

We use the notation $\text{Box}(123)$ throughout, as it is more convenient in longer formulas.

For a permutation $\pi = \pi_1 \cdots \pi_n \in \mathfrak{S}_n$, its *reverse*, *complement*, and *inverse* are defined, respectively, by $\pi_i^r = \pi_{n+1-i}$, $\pi_i^c = n+1 - \pi_i$, $\pi_i^{-1} = i$ for $1 \leq i \leq n$. Geometrically, these operations reflect the plot in a vertical line, reflect it in a horizontal line, and reflect it in the main diagonal, respectively. They act on mesh patterns by applying the same reflection to both the underlying permutation and the shaded cells. In particular, if $P = (\tau, R)$ has length k , then

$$\begin{aligned} P^r &= (\tau^r, \{(k-a, b) : (a, b) \in R\}), \\ P^c &= (\tau^c, \{(a, k-b) : (a, b) \in R\}), \\ P^{-1} &= (\tau^{-1}, \{(b, a) : (a, b) \in R\}). \end{aligned}$$

These symmetries preserve containment, avoidance, and coincidence. They also preserve boxed patterns: $\text{Box}(\tau)^r = \text{Box}(\tau^r)$, $\text{Box}(\tau)^c = \text{Box}(\tau^c)$, and $\text{Box}(\tau)^{-1} = \text{Box}(\tau^{-1})$.

If $X \subseteq [k-1]$, a *vincular pattern* (τ, X) requires $i_{x+1} = i_x + 1$ for every $x \in X$. We indicate bonds by underlining; thus 2413 is the pattern for which the entries playing 4 and 1 are adjacent. The notation 2-41-3 is equivalent. A *bivincular pattern* (τ, X, Y) additionally requires that the selected values of ranks y and $y+1$ be consecutive for each $y \in Y$ [9]. Vincular and bivincular patterns are mesh patterns: position bonds shade complete columns, and value bonds shade complete rows. For background on all of these families see Kitaev [20] and the survey of Steingrímsson [27].

For any pattern P , let $\text{Av}(P) = \bigcup_{n \geq 0} \text{Av}_n(P)$. Two patterns P and Q are *coincident*, written $P \equiv_c Q$, if $\text{Av}_n(P) = \text{Av}_n(Q)$ for every $n \geq 0$. Equivalently, every permutation contains P if and only if it contains Q . In informal terms, the two pattern-avoidance tests are indistinguishable even though the patterns may be drawn differently. This is stronger than *Wilf equivalence*, which asks only for $|\text{Av}_n(P)| = |\text{Av}_n(Q)|$ in every length.

We shall also use the obstruction to coincidence with classical patterns introduced by Tenner [29]. Write $G(\tau) = \{(i, \tau_i) : 1 \leq i \leq k\}$ for the graph of $\tau \in \mathfrak{S}_k$. A shaded square $(a, b) \in R$ is a *pointless square*, or an enclosed diagonal of length one, if its closed square has no vertex in $G(\tau)$. For $c \geq 1$, a set of shaded squares

$$D = \{(a + i, b + i) : 0 \leq i \leq c\} \subseteq R$$

is an *enclosed NE-diagonal* if

$$G(\tau) \cap \{(a + i, b + i) : 0 \leq i \leq c + 1\} = \{(a + i, b + i) : 1 \leq i \leq c\}.$$

Similarly, $D = \{(a + i, b - i) : 0 \leq i \leq c\} \subseteq R$ is an *enclosed SE-diagonal* if

$$G(\tau) \cap \{(a + i, b + 1 - i) : 0 \leq i \leq c + 1\} = \{(a + i, b + 1 - i) : 1 \leq i \leq c\}.$$

Either type has length $c + 1$ and is called *proper* when its length is greater than one. This is the square-based formulation in [13, Definition 3.1]. For a mesh pattern $P = (\tau, R)$, let $\text{enc}(P)$ denote its set of enclosed diagonals, regarded as sets of shaded squares.

Proposition 2.1 ([13], Lemmas 4.1 and 4.2). *If the mesh patterns $P = (\tau, R)$ and $Q = (\sigma, S)$ are coincident, then $\tau = \sigma$ and $\text{enc}(P) = \text{enc}(Q)$.*

Equality of enclosed diagonals is necessary but is not sufficient for general mesh-pattern coincidence [13]. We shall use the necessary direction explicitly in the proof of Theorem 5.2.

A mesh R is called *superfluous* for τ if its shading imposes no additional avoidance condition, that is, $(\tau, R) \equiv_c (\tau, \emptyset)$. Thus “superfluous” describes the whole mesh; it does not mean that its cells can necessarily be removed independently.

Theorem 2.2 ([29]). *The mesh R is superfluous for τ if and only if $\text{enc}((\tau, R)) = \emptyset$.*

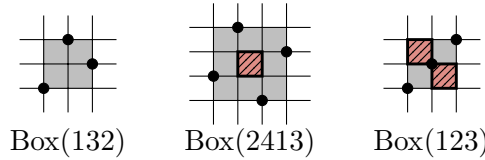


Figure 1: Enclosed diagonals in boxed meshes. Enclosed diagonals are drawn hatched and outlined as well as coloured, so that they remain distinguishable in grayscale. The left-hand mesh has none; the marked central square in the middle is a pointless enclosed diagonal; the two marked squares on the right form a proper SE-diagonal of length two.

Thus boxed patterns are *not* generally free of enclosed diagonals. Indeed, Tenner’s classification [29, Corollary 3.8] says that the only boxed patterns without one are those with underlying permutations 1, 12, 21, 132, 213, 231, and 312. In particular, every boxed pattern of length at least four has an enclosed diagonal.

3 Boxed patterns as interval-consecutive patterns

For an integer interval $I = [a, b]$, write $\pi|_I$ for the subword of π obtained by retaining entries whose values lie in I .

Definition 3.1. Let $\tau \in \mathfrak{S}_k$. An occurrence $\pi_{i_1} \cdots \pi_{i_k}$ of τ in π is a *boxed occurrence* if the open axis-parallel rectangle determined by its leftmost, rightmost, lowest and highest selected points contains no further point of the plot of π . The corresponding mesh pattern is denoted $\text{Box}(\tau)$.

Because a permutation plot has no repeated abscissae or ordinates, the condition can equivalently be stated with the closed rectangle: the closed bounding rectangle contains no plot point other than the selected ones.

Theorem 3.2. *For every $\tau \in \mathfrak{S}_k$ and every permutation π , the following are equivalent.*

- (i) π contains $\text{Box}(\tau)$;
- (ii) for some interval I of values, the word $\pi|_I$ has a consecutive factor whose standardization is τ .

Proof. Suppose first that points with positions $i_1 < \cdots < i_k$ form a boxed occurrence, and let u and v be the minimum and maximum of their values. No other point with value in $[u, v]$ can have position strictly between i_1 and i_k . Thus the selected entries form a consecutive factor of $\pi|_{[u, v]}$.

Conversely, suppose a consecutive factor of $\pi|_I$ standardizes to τ . Let u and v be the minimum and maximum values in this factor. Any extra point in its bounding rectangle would have value in $[u, v] \subseteq I$ and position between the first and last entries of the factor. It would therefore appear inside the factor of $\pi|_I$, a contradiction. $\square$

The criterion places boxed avoidance beside simsun-type restrictions, but with all value intervals rather than only initial intervals. It also gives a uniform way to search for occurrences: restrict to intervals, then use consecutive pattern matching. A second equivalent formulation, used in [3], is obtained by repeatedly deleting the leftmost, rightmost, smallest or largest entry: π contains $\text{Box}(\tau)$ if and only if some such sequence of external deletions ends at τ .

Remark 3.3. Deciding whether a given permutation contains a given boxed pattern has also been studied algorithmically. Bruner and Lackner [6] devote a section to boxed mesh patterns and give an $O(n^3)$ algorithm; Cho, Na and Sim [12] improve this to $O(n^2m)$ and then to $O(n^2 \log m)$, where m is the length of the pattern; and Amit, Bille, Hagge Cording, Li Gørtz and Wedel Vildhøj [4] give an $O(n^2)$ algorithm, which is optimal because the output can have size $\Omega(n^2)$.

The interval formulation also gives a useful inheritance principle. Say that $\sigma \in \mathfrak{S}_r$ is an *interval factor* of $\tau = \tau_1 \cdots \tau_k$ if, for some j and a ,

$$\{\tau_j, \tau_{j+1}, \dots, \tau_{j+r-1}\} = \{a, a+1, \dots, a+r-1\}$$

and $\sigma = \text{st}(\tau_j \cdots \tau_{j+r-1})$. Thus the entries forming σ occupy consecutive positions and consecutive ranks in τ .

Proposition 3.4. *If σ is an interval factor of τ , then every occurrence of $\text{Box}(\tau)$ contains an occurrence of $\text{Box}(\sigma)$. Consequently, $\text{Av}_n(\text{Box}(\sigma)) \subseteq \text{Av}_n(\text{Box}(\tau))$ for $n \geq 0$. In particular, failure of the Stanley–Wilf property for $\text{Box}(\sigma)$ implies failure of the property for $\text{Box}(\tau)$.*

Proof. Take the selected points of a boxed τ occurrence that correspond to the factor $\tau_j \cdots \tau_{j+r-1}$. No other selected point lies horizontally between the first and last of these points, because the positions of the factor are consecutive. No other selected point lies

vertically between their lowest and highest points, because their ranks are consecutive. An unselected point in the smaller bounding rectangle would also lie in the bounding rectangle of the original boxed occurrence. Hence the selected factor is a boxed σ occurrence. Taking contrapositives gives the class inclusion, and the final assertion follows by comparing cardinalities. $\square$

For example, Proposition 3.4 and the doubled-letter construction show immediately that $\text{Box}(\tau)$ has factorially many avoiders whenever τ has an interval factor 123 or 321. The “good pattern” argument of [3] handles many patterns without such a monotone interval factor and is therefore genuinely stronger than this inheritance principle.

4 The exceptional length-four patterns

Reverse, complement and inverse preserve boxed containment and avoidance. A direct calculation gives the two orbits in (2); in particular, inverse fixes 2143 and interchanges 2413 with 3142. The distinction is enumerative as well as formal:

n	0	1	2	3	4	5
$ \text{Av}_n(\text{Box}(2143)) $	1	1	2	6	23	106
$ \text{Av}_n(\text{Box}(2413)) $	1	1	2	6	23	104

Recall that an occurrence of the vincular pattern $\underline{2413}$ is a subsequence a, b, c, d in that positional order, with b and c adjacent and $c < a < d < b$.

Lemma 4.1. *A permutation contains the boxed pattern $\text{Box}(2413)$ if and only if it contains the vincular pattern $\underline{2413}$.*

Proof. Suppose a, b, c, d is a boxed 2413 occurrence, so $c < a < d < b$. Read the entries between b and c . Since the rectangle is empty, each is either above b or below c . Beginning with b and ending with c , there is an adjacent transition b', c' from the upper to the lower region. Since $b' \geq b$ and $c' \leq c$, $c < a < d < b$ gives $c' < a < d < b'$; and b' and c' lie weakly between b and c in position, so a precedes b' and d follows c' . Then a, b', c', d is an occurrence of $\underline{2413}$.

For the converse, use the external-deletion characterization in [3] discussed in Section 3. It is enough to show that whenever a permutation of length greater than four contains $\underline{2413}$, one of its four external entries can be deleted while preserving such an occurrence. Choose an occurrence a, b, c, d . If an external entry is not among these four, delete it; the bond survives, since b and c are adjacent and therefore no entry lies between them. The only remaining case is therefore that a, b, c, d are respectively the leftmost, largest, smallest and rightmost entries of the whole permutation; the assignment of roles is forced, because by $c < a < d < b$ the largest of the four is b and the smallest is c , while in positional order the first is a and the last is d . Let x be the entry immediately preceding b .

If $x > d$, deleting the largest entry b makes x adjacent to c , and a, x, c, d is still an occurrence. Suppose instead that $x < d$. If $x \neq a$, then $c < x < d < b$, so x, b, c, d is an occurrence and the leftmost entry a may be deleted. The only remaining possibility is $x = a$. Since a is the first entry and b and c are adjacent, the permutation begins with a, b, c . Moreover, d is the last entry, so, because the permutation has length greater than four, the entry y immediately following c exists and satisfies $y \neq d$. Since c and b are respectively the smallest and largest entries of the permutation, $c < y < b$. If $y > a$,

then $c < a < y < b$, so a, b, c, y is an occurrence and d may be deleted. If $y < a$, then $y < a < d < b$; deleting the smallest entry c makes b and y adjacent, and a, b, y, d is an occurrence. Thus an external entry can always be deleted while preserving an occurrence. Repeating the argument ends at the permutation 2413, proving boxed containment. $\square$

Remark 4.2. Contrary to what one might first expect, boxed meshes are not generally free of enclosed diagonals; see Figure 1. The two meshes in Lemma 4.1 have the same nonempty set of enclosed diagonals: the pointless square $(2, 2)$. Thus the enclosed-diagonal invariant of [13] is consistent with the coincidence, but Tenner’s superfluous-mesh theorem [29] does not prove it: that theorem applies precisely when there are no enclosed diagonals. Nor are the two meshes nested, so the one-square Shading Lemma [18, Lemma 3.11] cannot be applied directly. Its simultaneous extension [13, Section 7] is also a sufficient local shading criterion, not a converse that would establish this coincidence. Finally, the coincidence theorem for two vincular patterns does not apply because the boxed mesh is not vincular. The tightening argument above is therefore not an immediate consequence of these earlier results; it supplies the required global movement of the adjacency bond.

Theorem 4.3. *For every $n \geq 0$, $|\text{Av}_n(\text{Box}(2413))| = |\text{Av}_n(\text{Box}(3142))| = \text{SB}_n$, where SB_n is the n th semi-Baxter number (OEIS A117106). With $\text{SB}_0 = \text{SB}_1 = 1$, these numbers satisfy*

$$\text{SB}_n = \frac{11n^2 + 11n - 6}{(n+4)(n+3)} \text{SB}_{n-1} + \frac{(n-3)(n-2)}{(n+4)(n+3)} \text{SB}_{n-2} \quad (n \geq 2), \quad (3)$$

and, for $n \geq 2$,

$$\text{SB}_n = \frac{24}{(n-1)n^2(n+1)(n+2)} \sum_{j=0}^n \binom{n}{j+2} \binom{n+2}{j} \binom{n+j+2}{j+1}. \quad (4)$$

Proof. By Lemma 4.1, the avoiders of $\text{Box}(2413)$ are the semi-Baxter permutations, and hence have cardinality SB_n . Inverse symmetry gives the same enumeration for $\text{Box}(3142)$. The recurrence and sum are the semi-Baxter enumeration of Bouvel et al. [7, Proposition 13 and Theorem 14]. $\square$

In particular, $\text{SB}_n \sim A\mu^n n^{-6}$ with $\mu = ((\sqrt{5}-1)/2)^{-5}$ and an explicit positive constant A [7]. Boxed 2413 therefore satisfies the Stanley–Wilf property. It also follows that no injection from $\mathfrak{S}_{\lfloor n/2 \rfloor}$ to $\text{Av}_n(\text{Box}(2413))$ can exist for every n : such an injection would force a factorial lower bound on an exponentially bounded sequence.

Combining Theorem 4.3 with the factorial constructions in [3] leaves exactly one unresolved dihedral orbit at length four, namely $\{2143, 3412\}$.

The combination is worth recording as a single global statement. It also clarifies that coincidence and growth are separate questions: the noncoincidence theorem in Section 5 is not needed for the factorial lower bound in large lengths.

Theorem 4.4. *Let $\tau \in \mathfrak{S}_k$ with $k \geq 4$.*

- (i) *If $\tau \notin \{2143, 3412, 2413, 3142\}$, then $|\text{Av}_n(\text{Box}(\tau))| \geq \lfloor \frac{n}{2} \rfloor!$, so $\text{Box}(\tau)$ does not have the Stanley–Wilf property.*
- (ii) *If $\tau \in \{2413, 3142\}$, then $\text{Box}(\tau)$ has the Stanley–Wilf property and is enumerated by the semi-Baxter numbers.*
- (iii) *The Stanley–Wilf property remains open for $\tau \in \{2143, 3412\}$; these two cases are equivalent by complement.*

In particular, no boxed pattern of length at least five has the Stanley–Wilf property.

Proof. Part (i) is [3, Theorem 3]. Stirling's formula gives $(\lfloor \frac{n}{2} \rfloor!)^{1/n} \sim \sqrt{\frac{n}{2e}}$, which tends to infinity. Part (ii) is Theorem 4.3, and part (iii) follows from complement symmetry and the fact that neither of the preceding results applies to this orbit. For $k \geq 5$ none of the four exceptional permutations in part (i) has the required length, proving the last assertion. $\square$

Together with the short cases, Theorem 4.4 gives an almost complete classification over all boxed patterns. The Stanley–Wilf property is known to hold for the boxed patterns with underlying permutations 1, 12, 21, 132, 213, 231, 312, 2413, 3142; it is known to fail for every other boxed pattern except 2143 and 3412.

5 Classical, vincular and bivincular coincidences

We first record the classical case, which is now a consequence of the general coincidence theory for mesh patterns.

Proposition 5.1. *Let $\tau \in \mathfrak{S}_k$. Then $\text{Box}(\tau)$ is coincident with a classical pattern if and only if $k \leq 2$, or $k = 3$ and $\tau \in \{132, 213, 231, 312\}$. In every case the classical pattern is τ itself. In particular, no boxed pattern of length at least four is coincident with a classical pattern.*

Proof. By Proposition 2.1, a coincident classical pattern would have to be τ itself. By Theorem 2.2, coincidence with τ holds exactly when the full internal box has no enclosed diagonal. Applying that criterion gives precisely the displayed list; this is also [3, Proposition 1] and [29, Corollary 3.8]. $\square$

The answer changes for vincular patterns, but only in small lengths.

Theorem 5.2. *Let V be a vincular pattern of length k with at least one bond. Then $\text{Box}(\tau) \equiv_c V$ if and only if the pair occurs in the following list:*

$$\begin{array}{ll} k = 2 : & \text{Box}(12) \equiv_c \underline{12}, & \text{Box}(21) \equiv_c \underline{21}; \\ k = 3 : & \text{Box}(132) \equiv_c \underline{132}, & \text{Box}(213) \equiv_c \underline{213}, \\ & \text{Box}(231) \equiv_c \underline{231}, & \text{Box}(312) \equiv_c \underline{312}; \\ k = 4 : & \text{Box}(2413) \equiv_c \underline{2413}, & \text{Box}(3142) \equiv_c \underline{3142}. \end{array}$$

In particular, there is no boxed–vincular coincidence in length $k \geq 5$.

Proof. Write $V = (\sigma, X)$. If $\text{Box}(\tau) \equiv_c V$, the first necessary condition in Theorem 2.1 gives $\sigma = \tau$, while the second gives

$$\text{enc}(\text{Box}(\tau)) = \text{enc}(V). \quad (5)$$

We use this equality repeatedly.

Fix a bond $c \in X$. It shades the complete column c in V , including the two boundary cells $(c, 0)$ and (c, k) , neither of which is shaded in the boxed mesh. If $(c, 0)$ touched no point of $G(\tau)$, it would be a pointless enclosed diagonal of V that is absent from $\text{Box}(\tau)$, contradicting (5). It must therefore touch one of $(c, 1)$ and $(c + 1, 1)$, so one of τ_c, τ_{c+1} is 1. Applying the same argument to (c, k) shows that one of them is k . Hence

$$\{\tau_c, \tau_{c+1}\} = \{1, k\}. \quad (6)$$

Since the values 1 and k occur only once, two distinct bonds cannot both satisfy (6); consequently $X = \{c\}$.

Applying complement to both patterns if necessary, we may assume that $\tau_c = 1$ and $\tau_{c+1} = k$. Recall that V shades only column c . Every internal cell outside column c must therefore touch a point of $G(\tau)$; otherwise it would be a pointless enclosed diagonal of $\text{Box}(\tau)$ that is absent from V , contradicting (5).

Suppose first that $c \leq k - 2$, so that column $c + 1$ is an internal column. Its $k - 1$ internal cells can be touched only by the two plot points $(c + 1, \tau_{c+1}) = (c + 1, k)$ and $(c + 2, \tau_{c+2})$. The first point touches only one internal cell of column $c + 1$, while the second touches at most two. Thus at most three of the $k - 1$ internal cells in this column can touch $G(\tau)$. Since $k \geq 5$, at least one of these cells is pointless, a contradiction. Hence $c = k - 1$.

Similarly, if $c \geq 2$, consider the internal column $c - 1$. Its cells can be touched only by $(c - 1, \tau_{c-1})$ and $(c, \tau_c) = (c, 1)$. Again, the extreme-valued point touches only one internal cell and the other point touches at most two. For $k \geq 5$, this leaves a pointless cell in column $c - 1$, so $c = 1$.

We have obtained both $c = k - 1$ and $c = 1$, which would force $k = 2$. This contradicts $k \geq 5$, and therefore no boxed–vincular coincidence can occur in length at least five.

It remains to settle $k \leq 4$.

For $k = 2$, condition (6) leaves only 12 and 21. A permutation contains a classical 12 if and only if it has an adjacent ascent: a permutation with no adjacent ascent is decreasing. Hence $12 \equiv_c \underline{12}$, and similarly $21 \equiv_c \underline{21}$. Combining these observations with Theorem 5.1 gives the two asserted boxed coincidences.

For $k = 3$, (6) leaves exactly the four vincular patterns displayed in the theorem. In Claesson’s generalized-pattern notation, Lemma 21 of [11] proves $(b-a-c) \equiv_c (b-ac)$, that is, $213 \equiv_c \underline{213}$.

Reverse and complement give the other three identities. Each of the four underlying classical patterns is coincident with its boxed version by Theorem 5.1, proving all four length-three cases.

Finally let $k = 4$. Equation (6) gives twelve candidates. The following six are impossible; the indicated cell is pointless and shaded in the boxed mesh, but is not shaded in the vincular mesh, contradicting (5).

vincular pattern	cell	vincular pattern	cell
<u>1432</u>	(2, 1)	<u>4123</u>	(2, 3)
<u>2143</u>	(1, 3)	<u>3412</u>	(1, 1)
<u>2341</u>	(2, 1)	<u>3214</u>	(2, 3)

The six candidates not eliminated in this way are 1423, 2314, 2413, 3142, 3241, and 4132.

The remaining four noncoincidences have the following explicit certificates. In the last column we list every occurrence of the underlying classical pattern in the separator; the entry following the semicolon is the omitted fifth entry, which lies in its bounding rectangle and therefore blocks it from being boxed.

vincular pattern	separator	vincular occurrence	boxed candidates; blockers
<u>1423</u>	15324	1534	1534; 2 1524; 3
<u>2314</u>	24315	2415	2415; 3 2315; 4
<u>3241</u>	42351	4251	4251; 3 4351; 2
<u>4132</u>	51342	5132	5132; 4 5142; 3

Thus each separator contains the indicated vincular pattern and avoids the corresponding boxed pattern. The third candidate is the coincidence proved in Lemma 4.1, and the fourth is its reversal. This completes the classification. $\square$

There is also a natural coincidence with a genuinely bivincular pattern. It does not merely repackage the mesh as a set of position bonds.

Corollary 5.3. *We have $\text{Box}(2413) \equiv_c 2\underline{4}13 \equiv_c (2413, \{2\}, \{2\})$.*

Proof. By Lemma 4.1, it is enough to show that among occurrences a, b, c, d of $2\underline{4}13$ ($c < a < d < b$ and b and c are adjacent) there is one satisfying $d = a + 1$. Choose such an occurrence a, b, c, d minimizing $d - a$. If $a < e < d$ for some entry e of the permutation, then, because b and c are adjacent, e lies either to the left of b or to the right of c . In the first case e, b, c, d is a new occurrence with a smaller gap; in the second, a, b, c, e is. Both contradict minimality. Hence $d = a + 1$. The reverse implication is immediate. $\square$

The three coincident meshes in Corollary 5.3 are shown in Figure 2. The last has both a complete shaded column and a complete shaded row, so it is genuinely bivincular.

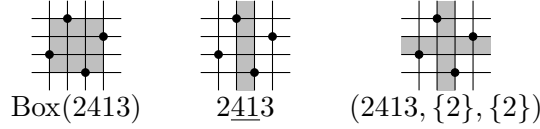


Figure 2: A boxed, a vincular and a genuinely bivincular presentation of the same containment condition.

Theorem 5.4. *Let $\tau \in \mathfrak{S}_k$, and let B be a bivincular pattern of length k . If $\text{Box}(\tau) \equiv_c B$, then $k \leq 4$. In particular, boxed–bivincular coincidences do not occur in arbitrarily large lengths.*

Proof. Write $B = (\sigma, X, Y)$, where $X, Y \subseteq [k - 1]$ are respectively its sets of position and value bonds. By Proposition 2.1, coincidence implies

$$\sigma = \tau \quad \text{and} \quad \text{enc}(\text{Box}(\tau)) = \text{enc}(B). \quad (7)$$

If $X = Y = \emptyset$, the result follows from Proposition 5.1. If exactly one of X, Y is nonempty, it follows from Theorem 5.2, after applying inversion when the bonds are value bonds. We may therefore assume that X and Y are both nonempty.

Fix $c \in X$. The complete shaded column c of B contains the boundary squares $(c, 0)$ and (c, k) , neither of which is shaded in the boxed mesh. Equality in (7) implies that neither boundary square can be pointless. The first must touch a plot point of value 1, and the second a plot point of value k . Hence $\{\tau_c, \tau_{c+1}\} = \{1, k\}$. The values 1 and k occur only once, so $X = \{c\}$. Applying the same argument after inversion gives, for the unique $d \in Y$,

$$\{\tau^{-1}(d), \tau^{-1}(d + 1)\} = \{1, k\}, \quad Y = \{d\}. \quad (8)$$

Thus the mesh of B consists of one complete column c and one complete row d .

Every internal square outside row d and column c must touch a point of $G(\tau)$. Otherwise it would be a pointless enclosed diagonal of $\text{Box}(\tau)$ absent from B , contradicting (7).

Suppose first that $k \geq 6$ and $c \leq k - 2$. Apart from the square in row d , internal column $c + 1$ contains $k - 2 \geq 4$ squares that must touch $G(\tau)$. They can touch only the plot

points in positions $c + 1$ and $c + 2$. By $\{\tau_c, \tau_{c+1}\} = \{1, k\}$, the first of these has value 1 or k and touches only one internal square of the column; the second touches at most two. This is impossible, so $c = k - 1$. The analogous argument in column $c - 1$, when $c \geq 2$, gives $c = 1$, a contradiction. Hence $k \leq 5$.

It remains to exclude $k = 5$. Complementing if necessary, assume $\tau_c = 1$ and $\tau_{c+1} = 5$.

If $c \leq 3$, the three squares of column $c + 1$ outside row d must all be touched. The point $(c + 1, 5)$ touches only row 4, while the point in position $c + 2$, writing $q = \tau_{c+2}$, touches rows $q - 1$ and q when these are internal. For the touched rows to contain $[4] \setminus \{d\}$, either $q = 2$ and $d = 3$, or $q = 3$ and $d = 1$; in particular, $d \in \{1, 3\}$. Similarly, if $c \geq 2$, column $c - 1$ is touched by $(c, 1)$ in row 1 and by the point in position $c - 1$. If $q = \tau_{c-1}$, the available rows are $\{1, q - 1, q\} \cap [4]$. Covering the three required squares forces $d \in \{2, 4\}$. Therefore c cannot lie in $\{2, 3\}$, and so $c \in \{1, 4\}$.

After reversing and, if necessary, complementing, we may take $c = 1$, $\tau_1 = 1$, and $\tau_2 = 5$. Equation (8) says that the values $d, d + 1$ occupy the first and last positions. It follows that $d = 1$ and $\tau_5 = 2$. The preceding column-2 argument then forces $\tau_3 = 3$, and hence $\tau = 15342$.

The mesh of B now contains the two boundary squares $D = \{(0, 1), (1, 0)\}$. They form a proper enclosed SE-diagonal surrounding the plot point $(1, 1)$. Neither square is shaded in Box(15342), so D is absent from its set of enclosed diagonals. This contradicts (7) and rules out $k = 5$. $\square$

6 The boxed mesh pattern 123

No closed formula is known for the number of permutations avoiding Box(123). In this section we give an exact maximum-insertion identity, use it to extend the known sequence, and prove a subfactorial upper bound. We then obtain a pattern-independent first-moment formula; in particular, the mean number of Box(123) occurrences is $(n + 2)H_n - 3n \sim n \log n$, where H_n is the n -th harmonic number.

6.1 Exact enumeration

Let $\mathcal{A}_n = \text{Av}_n(\text{Box}(123))$ and $a_n = |\mathcal{A}_n|$. For a word $w = w_1 \cdots w_t$ with distinct entries, set $R_i(w) = |\{j : i < j \leq t, w_j > w_i\}|$ and $\mathcal{Q}_t = \{w \in \mathfrak{S}_t : R_i(w) \neq 1 \text{ for all } i\}$. We call the gap after the first t entries of π *active* if insertion of the new maximum in that gap preserves boxed 123 avoidance.

Theorem 6.1. *Let $\pi \in \mathcal{A}_n$, and let $w = \pi_1 \cdots \pi_t$. The gap following w is active if and only if $\text{st}(w) \in \mathcal{Q}_t$. Moreover, $|\mathcal{Q}_0| = 1$ and $|\mathcal{Q}_t| = (t - 1)!$ for $t \geq 1$. Consequently,*

$$a_{n+1} = \sum_{\pi \in \mathcal{A}_n} \sum_{t=0}^n \mathbf{1}[\text{st}(\pi_1 \cdots \pi_t) \in \mathcal{Q}_t]. \quad (9)$$

Here $\mathbf{1}[\mathcal{E}]$ denotes the indicator of a statement $\mathcal{E}$: it equals 1 when $\mathcal{E}$ is true and 0 otherwise.

Proof. Any new boxed 123 occurrence must use the inserted maximum as its last and largest selected entry. If its first selected entry is w_i , the boxed condition says that exactly one entry to the right of w_i in the prefix w is larger than w_i : this unique entry is the middle selected point. This condition is also sufficient, since every other entry between w_i and the new maximum is then below w_i and hence outside the bounding rectangle. Thus the gap is inactive precisely when some $R_i(w)$ equals one.

For the enumeration, recall that the *Lehmer code* of $w = w_1 \cdots w_t \in \mathfrak{S}_t$ is

$$\ell_i(w) = |\{j : i < j \leq t, w_j < w_i\}|, \quad 1 \leq i \leq t.$$

The map $w \mapsto (\ell_1(w), \dots, \ell_t(w))$ is a bijection between $\mathfrak{S}_t$ and the set of vectors satisfying $0 \leq \ell_i \leq t - i$. Indeed, the permutation can be reconstructed successively by choosing w_i to be the $(\ell_i + 1)$ -st smallest of the values not yet used.

Since every entry to the right of w_i is either smaller or larger than w_i , we have $R_i(w) = t - i - \ell_i(w)$. Thus $(R_1(w), \dots, R_t(w))$, which we call the *complementary Lehmer code*, is also a bijective code in which the coordinates may independently take the values

$$R_i(w) \in \{0, 1, \dots, t - i\}.$$

For $i < t$, excluding the value 1 leaves $t - i$ choices, while $R_t(w) = 0$ has one choice. Therefore $|\mathcal{Q}_t| = \prod_{i=1}^{t-1} (t - i) = (t - 1)!$. $\square$

Iterating (9) by maximum insertion provides an efficient recursive generation and gives

$$\begin{aligned} (a_n)_{n=0}^{17} = & 1, 1, 2, 5, 15, 51, 194, 810, 3675, 17935, \\ & 93481, 517129, 3021133, 18559966, 119468349, \\ & 803213656, 5624984106, 40932781511. \end{aligned} \quad (10)$$

The terms through $n = 12$ agree with OEIS entry A201168 [26], which listed no further terms; the five terms $a_{13}, \dots, a_{17}$ are computed here. In particular $a_6 = 194$, which differs from the value recorded in [3]; we have verified $a_6 = 194$ by exhaustive inspection of all 720 permutations of length six.

Finding a closed coefficient formula for the sequence in (10) remains an open problem. One possible route is to refine $\mathcal{A}_n$ by the set of prefix lengths t for which the complementary Lehmer code avoids the digit one. Theorem 6.1 identifies exactly the statistic that a successful refinement must track.

6.2 A subfactorial upper bound

The factorial lower bound for a_n rules out the Stanley–Wilf property, but it does not determine how close the avoidance sequence is to $n!$. The next result gives a genuinely subfactorial upper bound.

Theorem 6.2. *Let $0 < x \leq 1$ satisfy*

$$2x + \sqrt{x} \leq x(1 + \sqrt{x})^2. \quad (11)$$

Then, for every $n \geq 1$,

$$a_n \leq 2^{5n} n^{n \log_2(1 + \sqrt{x})}. \quad (12)$$

The smallest admissible value is $x^ = (2 \cos \frac{\pi}{7} - 1)^2$, and hence*

$$a_n \leq 2^{5n} n^{\beta n}, \quad \beta = \log_2 \left(2 \cos \frac{\pi}{7} \right) \approx 0.8496. \quad (13)$$

Consequently,

$$\frac{1}{2} \leq \liminf_{n \rightarrow \infty} \frac{\log a_n}{n \log n} \leq \limsup_{n \rightarrow \infty} \frac{\log a_n}{n \log n} \leq \beta < 1. \quad (14)$$

Here the lower bound follows from the doubled-letter injection recalled in the introduction, together with the fact that a_n is nondecreasing. Indeed, inserting a new maximum in front of an avoider creates no boxed 123 occurrence, because the new entry has no entry to its left and can only play the last selected point; hence $a_n \geq a_{2\lfloor n/2 \rfloor} \geq \lfloor n/2 \rfloor!$ for every n . We call $\limsup_{n \rightarrow \infty} \log a_n / (n \log n)$ the *growth exponent* of $\text{Av}(\text{Box}(123))$; by (14) it lies between $\frac{1}{2}$ and β . The upper bound also shows that a uniformly random permutation avoids $\text{Box}(123)$ with probability at most $n^{-(1-\beta)n+O(n/\log n)}$.

To prove the theorem, for a permutation π let $\text{asc}(\pi)$ be its number of ascents, and, for $0 < x \leq 1$, set

$$Z_n(x) = \sum_{\pi \in \mathcal{A}_n} x^{\text{asc}(\pi)}, \quad \tau(x) = \frac{2x + \sqrt{x}}{(1 + \sqrt{x})^2}.$$

For $0 < x \leq 1$ one has $0 < \tau(x) \leq 1$.

Lemma 6.3. *For $n \geq 2$, with $h_1 = \lfloor n/2 \rfloor$ and $h_2 = \lceil n/2 \rceil$, one has*

$$Z_n(x) \leq \frac{2(1 + \sqrt{x})^{n-1}}{\tau(x)} Z_{h_1}(\tau(x)) Z_{h_2}(\tau(x)). \quad (15)$$

Proof. For $\pi \in \mathfrak{S}_n$, call the entries with values at most h_1 *low* and the remaining entries *high*. Let $L \in \mathfrak{S}_{h_1}$ and $H \in \mathfrak{S}_{h_2}$ be the patterns of the low and high subsequences. Number the gaps around the low entries from 0 to h_1 , and number the slots between consecutive high entries from 1 to $h_2 - 1$. Define

$$U = \{i : \text{low gap } i \text{ contains a high entry}\}, \\ D = \{j : \text{high slot } j \text{ contains a low entry}\}.$$

The nonempty low gaps contain the maximal blocks of high entries. Thus $|D| = |U| - 1$, and π is recovered from (L, H, U, D) by cutting H at the slots in D and placing the resulting blocks in the gaps in U .

Put $T = |U \setminus \{0\}|$. This is the number of adjacent low–high pairs, and $T \geq |D|$. Let I_L be the number of ascents i of L for which $i \notin U$, and let I_H be the number of ascents j of H for which $j \notin D$. For $\pi \in \mathcal{A}_n$ the following elementary restrictions hold:

- (i) $L \in \mathcal{A}_{h_1}$ and $H \in \mathcal{A}_{h_2}$;
- (ii) neither L nor H has two consecutive ascents;
- (iii) if i is an ascent of L , then it is not the case that $i \notin U$ and $i + 1 \in U$;
- (iv) if $j \geq 2$ is an ascent of H , then it is not the case that $j - 1 \in D$ and $j \notin D$.

For (i), a point inside the rectangle of an occurrence among the low entries would itself be low, and similarly for the high entries. For (ii), three entries occupying consecutive positions of L and increasing in value form a boxed occurrence in π : between them there are no other low entries, while all intervening high entries lie above their vertical range. The argument for H is analogous. In (iii) the entries L_i, L_{i+1} followed by the first high entry in gap $i + 1$ occupy three consecutive positions of π and increase; in (iv), use the last low entry of slot $j - 1$ followed by H_j, H_{j+1} . These triples would be boxed occurrences, proving the restrictions.

Every adjacent ascent of π is either low–high, low–low, or high–high, and therefore

$$\text{asc}(\pi) = T + I_L + I_H. \quad (16)$$

Let E_L be the set of ascents of L , and let E_H be the set of ascents $j \geq 2$ of H . From (16), $T \geq |D|$, and $0 < x \leq 1$, we obtain $x^{\text{asc}(\pi)} \leq f(U)g(D)$, where

$$f(U) = x^{|U \setminus \{0\}|/2} x^{|\{i \in E_L : i \notin U\}|}, \quad g(D) = x^{|D|/2} x^{|\{j \in E_H : j \notin D\}|}.$$

If 1 is an ascent of H , its contribution has been omitted from g ; since $x \leq 1$, this only enlarges the right-hand side.

By (ii), the pairs $\{i, i+1\}$ with $i \in E_L$ are disjoint. In summing $f(U)$ over all subsets $U \subseteq \{0, 1, \dots, h_1\}$ allowed by (iii), the index 0 contributes 2, each index in no such pair contributes $1 + \sqrt{x}$, and each pair contributes $x + \sqrt{x} + x = 2x + \sqrt{x}$; the fourth membership pattern is excluded by (iii). Hence

$$\sum_U f(U) = 2(1 + \sqrt{x})^{h_1} \tau(x)^{|E_L|}.$$

The same calculation, carried out over all subsets $D \subseteq \{1, \dots, h_2 - 1\}$ allowed by (iv), gives

$$\sum_D g(D) = (1 + \sqrt{x})^{h_2-1} \tau(x)^{|E_H|}.$$

The encoding by (L, H, U, D) is injective, and every $\pi \in \mathcal{A}_n$ yields a pair (U, D) allowed by (iii) and (iv) and satisfying $|D| = |U| - 1$. Dropping that last relation enlarges the set of encodings summed over, so summing over all allowed pairs (U, D) gives

$$\begin{aligned} Z_n(x) &\leq 2(1 + \sqrt{x})^{n-1} \sum_{L \in \mathcal{A}_{h_1}} \sum_{H \in \mathcal{A}_{h_2}} \tau(x)^{|E_L|+|E_H|} \\ &\leq \frac{2(1 + \sqrt{x})^{n-1}}{\tau(x)} Z_{h_1}(\tau(x)) Z_{h_2}(\tau(x)), \end{aligned}$$

because $|E_L| + |E_H| \geq \text{asc}(L) + \text{asc}(H) - 1$ and $\tau(x) \leq 1$. $\square$

Proof of Theorem 6.2. Condition (11) is precisely $\tau(x) \leq x$. Since $Z_h(y)$ is increasing in y , Lemma 6.3 yields

$$Z_n(x) \leq \frac{2}{\tau(x)} (1 + \sqrt{x})^{n-1} Z_{h_1}(x) Z_{h_2}(x).$$

Put

$$F(n) = \log_2 Z_n(x), \quad \gamma_x = \log_2(1 + \sqrt{x}), \quad c = \log_2 \frac{2}{x^*} < 1.64.$$

The function τ is increasing on $(0, 1]$, so $\tau(x) \geq \tau(x^*) = x^*$ for admissible x . Hence $F(1) = 0$ and $F(n) \leq \gamma_x n + c + F(h_1) + F(h_2)$. Induction gives

$$F(n) \leq \gamma_x n \lceil \log_2 n \rceil + c(2n - 1).$$

Indeed, both h_i are at most $2^{\lceil \log_2 n \rceil - 1}$, so substituting the two inductive bounds cancels one halving level and gives the displayed inequality.

Since $\text{asc}(\pi) \leq n - 1$ and $x \leq 1$, we have $a_n \leq x^{-n} Z_n(x)$. Using $\lceil \log_2 n \rceil \leq \log_2 n + 1$, $\gamma_x \leq 1$, and $\log_2(1/x^*) < 0.64$, we obtain

$$\log_2 a_n \leq \gamma_x n \log_2 n + (\gamma_x + 3.28 + 0.64)n \leq \gamma_x n \log_2 n + 5n,$$

which proves (12).

Finally, with $s = \sqrt{x}$, condition (11) becomes $s^3 + 2s^2 - s - 1 \geq 0$. On $0 < s \leq 1$ its unique threshold is $s^* = 2 \cos(\pi/7) - 1$: after setting $t = 1 + s$, the corresponding equation is $t^3 - t^2 - 2t + 1 = 0$. Its three roots are $2 \cos(\pi/7)$, $2 \cos(3\pi/7)$, and $2 \cos(5\pi/7)$, of which only the first lies in $(1, 2]$. That s^* is the only sign change on $(0, 1]$ follows from a direct inspection: writing $p(s) = s^3 + 2s^2 - s - 1$, the derivative $p'(s) = 3s^2 + 4s - 1$ vanishes only at $s = (\sqrt{7} - 2)/3 = 0.2153\dots$, so p decreases on $(0, (\sqrt{7} - 2)/3)$ and increases on $((\sqrt{7} - 2)/3, 1]$; since $p(0) = -1 < 0 < 1 = p(1)$, the polynomial p is negative on $(0, s^*)$ and nonnegative on $[s^*, 1]$. Thus $x^* = (s^*)^2$ and $1 + \sqrt{x^*} = 2 \cos(\pi/7)$, proving (13). $\square$

6.3 Average occurrence counts for boxed patterns

For $\tau \in \mathfrak{S}_k$ define

$$F_{n,\tau}(q) = \sum_{\pi \in \mathfrak{S}_n} q^{N_{\text{Box}(\tau)}(\pi)}, \quad (17)$$

where $N_{\text{Box}(\tau)}(\pi)$ is the number of occurrences of $\text{Box}(\tau)$ in π . Its constant term is the avoidance number: $F_{n,\tau}(0) = |\text{Av}_n(\text{Box}(\tau))|$. For uniformly random $\pi \in \mathfrak{S}_n$, its derivative satisfies

$$\frac{F'_{n,\tau}(1)}{n!} = \mathbb{E}[N_{\text{Box}(\tau)}(\pi)]. \quad (18)$$

The full occurrence distribution for $\text{Box}(123)$ is open, but the first moment has a simple closed form. In fact, it is independent of the choice of the underlying pattern of a fixed length.

Proposition 6.4. *Let $k \geq 2$, let $\tau \in \mathfrak{S}_k$, and write $N_{\text{Box}(\tau)}(\pi)$ as in (17). If π is uniformly distributed over $\mathfrak{S}_n$, where $n \geq k$, then*

$$\mathbb{E}[N_{\text{Box}(\tau)}(\pi)] = \frac{(n+k-1)(H_n - H_{k-2}) - 2(n-k+2)}{(k-2)!}, \quad (19)$$

where $H_m = \sum_{j=1}^m 1/j$ is the m -th harmonic number and $H_0 = 0$. In particular,

$$\mathbb{E}[N_{\text{Box}(123)}(\pi)] = (n+2)H_n - 3n. \quad (20)$$

Moreover, for fixed n and k , the expectation in (19) is independent of the choice of $\tau \in \mathfrak{S}_k$: all boxed patterns of length k have the same first moment on $\mathfrak{S}_n$.

Proof. Fix selected positions $I = \{i_1 < \dots < i_k\}$ and put $d = i_k - i_1$. The block from i_1 to i_k has length $d+1$, and its relative order is uniformly distributed over $\mathfrak{S}_{d+1}$. The selected entries form an occurrence of $\text{Box}(\tau)$ exactly when they receive k consecutive ranks in this block, arranged in the order τ . There are $d-k+2$ choices for this interval of ranks, after which the other $d-k+1$ ranks may be arranged arbitrarily. Consequently,

$$\Pr(I \text{ is a } \text{Box}(\tau)\text{-occurrence}) = \frac{(d-k+2)!}{(d+1)!}.$$

For a fixed span d , there are $(n-d)\binom{d-1}{k-2}$ possible sets I . Linearity of expectation therefore gives

$$\begin{aligned} \mathbb{E}[N_{\text{Box}(\tau)}(\pi)] &= \sum_{d=k-1}^{n-1} (n-d) \binom{d-1}{k-2} \frac{(d-k+2)!}{(d+1)!} \\ &= \frac{1}{(k-2)!} \sum_{d=k-1}^{n-1} \frac{(n-d)(d-k+2)}{d(d+1)} \\ &= \frac{(n+k-1)(H_n - H_{k-2}) - 2(n-k+2)}{(k-2)!}. \end{aligned}$$

Setting $k = 3$ proves (20). □

Remark 6.5. Since $H_n = \log n + \gamma + O(1/n)$, where γ is the Euler–Mascheroni constant, (20) gives $\mathbb{E}[N_{\text{Box}(123)}(\pi)] = n \log n + (\gamma - 3)n + O(\log n)$. Thus a uniformly random permutation of length n has $n \log n + O(n)$ boxed 123 occurrences on average. This identifies the natural scale of the statistic, although a first-moment estimate alone does not imply concentration or determine the probability of avoidance.

7 The remaining orbit $\{2143, 3412\}$

There is a useful one-sided analogue of the tightening lemma from Section 4. Recall that an occurrence of $2\underline{1}43$ has the form a, x, y, d in positional order, where x and y are adjacent and $x < a < d < y$.

Lemma 7.1. *Every permutation containing $\text{Box}(2143)$ contains $2\underline{1}43$. Equivalently,*

$$\text{Av}_n(2\underline{1}43) \subseteq \text{Av}_n(\text{Box}(2143)) \quad (n \geq 0).$$

Proof. Let a, b, c, d be a boxed 2143 occurrence in positional order, so $b < a < d < c$. Every entry strictly between b and c in position lies either below b or above c ; an entry with value strictly between them would lie in the empty bounding rectangle. Read the permutation from b to c , regarding b as belonging to the lower region and c as belonging to the upper region. At some point there is an adjacent transition x, y from the lower region to the upper region. Hence $x \leq b < a < d < c \leq y$. The inequalities relevant to the four distinct entries a, x, y, d are strict, so these entries form an occurrence of $2\underline{1}43$. The avoidance-class inclusion is the contrapositive. $\square$

Remark 7.2. The reverse implication in Lemma 7.1 is false. For example, 24153 contains $2\underline{1}43$ in the entries $2, 1, 5, 3$, but avoids $\text{Box}(2143)$: the intervening entry 4 obstructs the only candidate. This is also forced by the classification in Theorem 5.2. Moreover, $2\underline{1}43$ -avoiders are the plane permutations and are enumerated by the semi-Baxter numbers [7]. Thus the lemma gives only the exponential lower bound $\text{SB}_n \leq |\text{Av}_n(\text{Box}(2143))|$. The inclusion has the wrong direction for an exponential upper bound and is far too small to yield a factorial lower bound. Its value is structural: it identifies the semi-Baxter class as a canonical subclass of the remaining boxed avoidance class and explains why tightening alone cannot resolve the Stanley–Wilf question.

We next observe that an injection from $\mathfrak{S}_m$ into avoiders whose length is an affine function of m is sufficient to disprove the Stanley–Wilf property. The observation applies to any mesh pattern, not only to boxed ones.

Proposition 7.3. *Let P be a fixed mesh pattern. Suppose that there is a constant integer $c \geq 1$, an integer r , and, for all sufficiently large m , an injection*

$$\Phi_m : \mathfrak{S}_m \longrightarrow \text{Av}_{cm+r}(P),$$

where $cm + r \geq 0$. Then P does not have the Stanley–Wilf property.

Proof. Writing $A_n = |\text{Av}_n(P)|$, injectivity gives $A_{cm+r} \geq m!$. Hence, by Stirling’s formula,

$$A_{cm+r}^{1/(cm+r)} \geq (m!)^{1/(cm+r)} \sim \left(\frac{m}{e}\right)^{1/c} \longrightarrow \infty.$$

Thus $\limsup_{n \rightarrow \infty} A_n^{1/n} = \infty$. $\square$

Consequently there is no need to insist on doubling: an embedding into permutations of length $2m - 1$, $3m$, $4m$, or, more generally, $cm + r$ for fixed c and r would be equally decisive. Increasing c may provide more room for blockers, although the extra entries can also create new occurrences.

Here is a concrete format in which such an injection may exist. For $\pi = \pi_1 \cdots \pi_m \in \mathfrak{S}_m$ and $\rho = \rho_1 \cdots \rho_{m-1} \in \mathfrak{S}_{m-1}$, define the strict interleaving

$$J_\rho(\pi) = (2\pi_1 - 1, 2\rho_1, 2\pi_2 - 1, 2\rho_2, \dots, 2\rho_{m-1}, 2\pi_m - 1) \in \mathfrak{S}_{2m-1}. \quad (21)$$

For $m = 1$ the even word is empty. Thus the prescribed odd entries occur in their original order, and exactly one even entry is placed in each internal gap. Equivalently, one permutes the set $\{2, 4, \dots, 2m - 2\}$ between the fixed odd entries $2\pi_1 - 1, \dots, 2\pi_m - 1$.

The odd-position entries recover the input without any auxiliary data:

$$\pi_i = \frac{J_\rho(\pi)_{2i-1} + 1}{2}. \quad (22)$$

For example, if $\pi = 231$ and $\rho = 21$, then $J_{21}(231) = (3, 4, 5, 2, 1)$, which avoids $\text{Box}(2143)$, and its odd-position entries $3, 5, 1$ recover 231 by (22).

There is also a simple reason that formulations beginning with a forced global minimum can be shortened. If a permutation beginning with 1 avoids $\text{Box}(2143)$, then deleting this 1 and reducing all other entries by one preserves avoidance. Conversely, adjoining a new global minimum on the left preserves avoidance, since the minimum in the underlying pattern 2143 is its second entry and therefore cannot be played by the new leftmost entry. This is a special case of the general observation that deleting a global extremum cannot uncover a boxed occurrence: the deleted point lies outside the vertical bounding interval of every occurrence formed by the remaining entries.

Conjecture 7.4. Let $m \geq 1$. For every $\pi \in \mathfrak{S}_m$, there exists $\rho \in \mathfrak{S}_{m-1}$ such that $J_\rho(\pi)$ avoids $\text{Box}(2143)$.

The conjecture has been verified exhaustively for $m \leq 8$. If true, choose the lexicographically first admissible ρ ; equation (22) then gives an explicit injection

$$\mathfrak{S}_m \hookrightarrow \text{Av}_{2m-1}(\text{Box}(2143)), \quad \pi \longmapsto J_{\rho(\pi)}(\pi),$$

and hence, by Proposition 7.3, the factorial lower bound sought in [3]. Conjecture 7.4 is stronger than the growth conclusion that it would imply, since factorial growth might arise from a different construction. We therefore record the expected growth statement separately.

Conjecture 7.5. The avoidance sequence of $\text{Box}(2143)$ has factorial growth, and hence $\text{Box}(2143)$ does not have the Stanley–Wilf property.

The following occurrence-preserving doubling provides a first step toward the conjecture. For $\pi = \pi_1 \cdots \pi_m \in \mathfrak{S}_m$, define

$$\Delta(\pi) = (2\pi_1 - 1)(2\pi_1) (2\pi_2 - 1)(2\pi_2) \cdots (2\pi_m - 1)(2\pi_m). \quad (23)$$

Thus each entry is replaced by an adjacent increasing pair of consecutive values, and the entries in the odd positions recover π .

Proposition 7.6. For every $\pi \in \mathfrak{S}_m$,

$$N_{\text{Box}(2143)}(\Delta(\pi)) = N_{\text{Box}(2143)}(\pi). \quad (24)$$

In particular, π avoids $\text{Box}(2143)$ if and only if $\Delta(\pi)$ does.

Proof. Call the two entries replacing x the x -block. A 2143 occurrence in $\Delta(\pi)$ uses at most one entry from any block. Indeed, two entries from one block would be adjacent selected entries in increasing order. The only adjacent ascent in 2143 is from its second selected entry, the minimum, to its third, the maximum; two consecutive values cannot play these roles while leaving two further selected values strictly between them.

Write φ for the projection that sends a selected entry of the x -block to x ; it is well defined on the selected entries by the previous paragraph, since each block contributes at most one of them. It follows that the four selected blocks of any 2143 occurrence project under φ to four entries a, b, c, d of π , in positional order, with $b < a < d < c$. If the occurrence in $\Delta(\pi)$ is boxed, then the projected occurrence is boxed as well: any intervening entry x of π with $b < x < c$ would place both entries of its block inside the bounding rectangle in $\Delta(\pi)$.

Conversely, let a, b, c, d be a boxed 2143 occurrence of π . It has the unique boxed lift

$$2a, \quad 2b, \quad 2c - 1, \quad 2d - 1. \quad (25)$$

To see uniqueness, choosing $2a - 1$ would put its mate $2a$ immediately to the right inside the rectangle; choosing $2b - 1$ would put $2b$ inside; choosing $2c$ would put $2c - 1$ inside; and choosing $2d$ would put $2d - 1$ inside. The choices in (25) avoid these obstructions. Their unselected mates lie, respectively, to the left of the rectangle, below it, above it, and to its right. Any other point inside the lifted rectangle would belong to an intervening block x with $b < x < c$, contradicting that the original occurrence was boxed. Thus φ and this unique lift are mutually inverse maps between the boxed 2143 occurrences of $\Delta(\pi)$ and those of π , proving (24). $\square$

Thus increasing doubling itself creates no additional boxed 2143 occurrences. It supplies a canonical alternating extension of length $2m$ whenever the prescribed odd subsequence already avoids the pattern. This is not yet the strict length- $(2m - 1)$ completion required by Conjecture 7.4; for a general input, the remaining task is to arrange the even entries so as to destroy the projected occurrences without creating new ones.

8 Boxed 12 and strong Bruhat covers

We finish the structural part of the paper with a distributional observation about the smallest nontrivial boxed pattern, $\text{Box}(12) = \begin{smallmatrix} \downarrow & \downarrow \\ \uparrow & \uparrow \end{smallmatrix}$. The avoidance sequence of this pattern is immediate: for each n , the decreasing permutation $n(n - 1) \cdots 1$ is its unique avoider. However, to our knowledge no general formula for its distribution is known, either in the mesh-pattern literature or in the literature on the strong Bruhat order, where only the extremal values and the mean of the statistic appear; see Corollary 8.3. The statistic has a natural interpretation in a different area of algebraic combinatorics.

For $\pi \in \mathfrak{S}_n$, let $\ell(\pi) = |\{(i, j) : 1 \leq i < j \leq n, \pi_i > \pi_j\}|$ be its inversion number. If $1 \leq i < j \leq n$, let $\pi(i \leftrightarrow j)$ be obtained by interchanging the entries in positions i and j . The *strong Bruhat order* on $\mathfrak{S}_n$ is the transitive closure of the covering relations

$$\pi \lessdot \pi(i \leftrightarrow j) \iff \ell(\pi(i \leftrightarrow j)) = \ell(\pi) + 1.$$

The *up-degree* of π is $d_+(\pi) = |\{\sigma \in \mathfrak{S}_n : \pi \lessdot \sigma\}|$; the down-degree $d_-(\pi)$ is defined analogously by counting $\sigma \lessdot \pi$. See Adin and Roichman [1] for these statistics and Bouvel, Ferrari and Tenner [8] for descriptions of Bruhat covering relations in mesh-pattern language.

Let $b(\pi)$ denote the number of occurrences of $\text{Box}(12)$ in π .

Proposition 8.1. *For every permutation π , $b(\pi) = d_+(\pi)$.*

Proof. Choose $i < j$ with $a = \pi_i < \pi_j = b$. Interchanging a and b changes the inversion number by $1 + 2|\{k : i < k < j, a < \pi_k < b\}|$. It therefore gives an upward Bruhat cover exactly when no entry strictly between positions i and j has value strictly between a and b . This is precisely the empty-cell condition for the selected pair to be an occurrence of $\text{Box}(12)$. $\square$

Remark 8.2. Kitaev and Liese [21, Theorem 12] proved that b has Catalan distribution on $\text{Av}_n(132)$: for $0 \leq k \leq n-1$,

$$|\{\pi \in \text{Av}_n(132) : b(\pi) = k\}| = C(n-1, k) = \frac{n-k}{n} \binom{n-1+k}{n-1},$$

where $C(r, k)$ denotes the (r, k) entry of Catalan's triangle. In view of Theorem 8.1, this also gives the distribution of the strong Bruhat up-degree on the Catalan class $\text{Av}_n(132)$.

In the notation of (17), define its distribution polynomial by

$$D_n(q) = F_{n,12}(q) = \sum_{\pi \in \mathfrak{S}_n} q^{b(\pi)}.$$

Complementation interchanges up-degree and down-degree. Consequently the results of Adin and Roichman [1, Propositions 2.1 and 2.9, and Theorem 4.1] translate as follows.

Corollary 8.3. *For $n \geq 1$, we have*

$$\deg D_n(q) = \left\lfloor \frac{n^2}{4} \right\rfloor, \tag{26}$$

$$[q^{\lfloor n^2/4 \rfloor}] D_n(q) = \begin{cases} n, & n \text{ odd}, \\ n/2, & n \text{ even}, \end{cases} \tag{27}$$

$$\frac{D'_n(1)}{n!} = (n+1)H_n - 2n, \tag{28}$$

where $H_n = \sum_{j=1}^n 1/j$ is the n -th harmonic number.

Remark 8.4. The familiar stronger algebraic properties do not hold in general. Already

$$D_3(q) = 1 + 2q + 3q^2$$

has nonreal zeros. Moreover, for every $n \geq 3$ the constant coefficient is 1, whereas (27) shows that the leading coefficient is greater than 1. Thus $D_n(q)$ is not palindromic, ruling out γ -positivity in its usual palindromic sense. Log-concavity also fails: the polynomial $D_7(q)$ recorded in [28] has $[q^{10}]D_7(q) = 129$, $[q^{11}]D_7(q) = 26$, $[q^{12}]D_7(q) = 7$, and $26^2 < 129 \cdot 7$. On the other hand, exhaustive enumeration gives a unimodal coefficient sequence for every $D_n(q)$ with $n \leq 11$.

This evidence suggests that unimodality may be the appropriate surviving property.

Conjecture 8.5. For every $n \geq 0$, the coefficient sequence of $D_n(q)$ is unimodal.

Remark 8.6. Interestingly, harmonic numbers also describe the complete distribution of a rather different mesh pattern. Let

$$p = \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline \end{array}$$

be the border mesh pattern studied in [21]. If $p_{n,k}$ denotes the number of permutations in $\mathfrak{S}_n$ with exactly k occurrences of p , then [21, Theorems 2 and 3], for $n \geq 4$,

$$p_{n,k} = (n-2)!(H_{n-2} - H_k) \quad (1 \leq k \leq n-3),$$

whereas $p_{n,0} = (n-2)!(H_{n-2} + n^2 - 2n + 2)$.

Equation (28) is also the $k = 2$ case of Proposition 6.4. The following direct derivation refines it by the positional span of an occurrence. Fix positions $i < j$ and put $d = j - i$. In the block $\pi_i \cdots \pi_j$ the two endpoint ranks are a uniformly distributed ordered pair of distinct elements of $[d+1]$. They form a boxed 12 occurrence exactly when the right endpoint has rank one larger than the left endpoint. There are d such pairs out of $d(d+1)$, so

$$\Pr((i, j) \text{ is a Box}(12)\text{-occurrence}) = \frac{1}{d+1}.$$

Linearity of expectation gives

$$\mathbb{E}[b(\pi)] = \sum_{d=1}^{n-1} \frac{n-d}{d+1} = (n+1)H_n - 2n.$$

Equivalently, the exponential generating function for the total number of occurrences is

$$\sum_{n \geq 0} \frac{D'_n(1)}{n!} x^n = \frac{-\log(1-x) - x}{(1-x)^2}. \quad (29)$$

Maximum insertion gives an exact identity for the full distribution, although it does not close in the single sequence of polynomials $D_n(q)$. For a word $w = w_1 \cdots w_t$ with distinct entries, let $\text{rlmax}(w) = |\{i \in [t] : w_i > w_j \text{ for every } i < j \leq t\}|$ be its number of right-to-left maxima, and set $\text{rlmax}(\varepsilon) = 0$.

Proposition 8.7. *For $\pi \in \mathfrak{S}_n$ and $0 \leq t \leq n$, let $\pi^{(t)} = \pi_1 \cdots \pi_t (n+1) \pi_{t+1} \cdots \pi_n$. Then*

$$b(\pi^{(t)}) = b(\pi) + \text{rlmax}(\pi_1 \cdots \pi_t). \quad (30)$$

Consequently,

$$D_{n+1}(q) = \sum_{\pi \in \mathfrak{S}_n} q^{b(\pi)} \sum_{t=0}^n q^{\text{rlmax}(\pi_1 \cdots \pi_t)}. \quad (31)$$

Proof. Inserting the new maximum neither creates nor destroys an occurrence using only old entries. A new occurrence must end at $n+1$. Its first entry π_i , where $i \leq t$, forms a boxed 12 occurrence with $n+1$ exactly when no later entry of the prefix $\pi_1 \cdots \pi_t$ is larger than π_i , that is, exactly when π_i is a right-to-left maximum of the prefix. This proves (30); summing over the $n+1$ insertion sites and all parents proves (31). $\square$

Formula (31) shows both the promise and the difficulty of maximum insertion: a closed recurrence must retain the joint distribution of $b(\pi)$ with the right-to-left-maximum counts of all prefixes. We do not know a finite catalytic refinement that accomplishes this, and the full polynomial $D_n(q)$ remains open.

9 Concluding directions

The growth classification is complete except for one length-four orbit. Theorem 4.4, together with the factorial construction of [3], shows that no boxed pattern of length at least five has the Stanley–Wilf property. In length four, twenty patterns have factorial lower bounds, the orbit $\{2413, 3142\}$ has semi-Baxter and hence exponential growth, and only $\{2143, 3412\}$ remains unresolved. We conjecture that this orbit has factorial growth; see Conjecture 7.5.

The coincidence results also reveal a sharp length boundary. Theorem 5.2 shows that boxed–vincular coincidences stop in length four, while Theorem 5.4 shows that the genuinely bivincular coincidence in Corollary 5.3 is also maximal in length. The interval-consecutive criterion provides a complementary structural viewpoint: boxed τ avoidance is consecutive τ avoidance on every interval of values. Moreover, Proposition 3.4 transfers factorial lower bounds from interval factors to larger boxed patterns.

For boxed 123, Theorem 6.1 gives an exact maximum-insertion identity and five new terms, while Theorem 6.2 bounds its growth exponent by a constant below 0.85. A closed enumeration remains open. Proposition 6.4 also shows that all boxed patterns of a fixed length have the same first moment; in particular, the expected number of boxed 123 occurrences is asymptotic to $n \log n$.

Finally, for boxed 12, Proposition 8.1 identifies the occurrence statistic with up-degree in the strong Bruhat order. Although real-rootedness, γ -positivity and log-concavity fail in small lengths, the available data motivate Conjecture 8.5, which asserts unimodality of the distribution polynomials.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used GPT (OpenAI) in order to assist with language editing, exploratory testing of conjectures, proof development, and the drafting of parts of the paper. After using this tool, the authors reviewed, corrected, and verified the resulting text, arguments, and references, and take full responsibility for the content of the manuscript.

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